

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

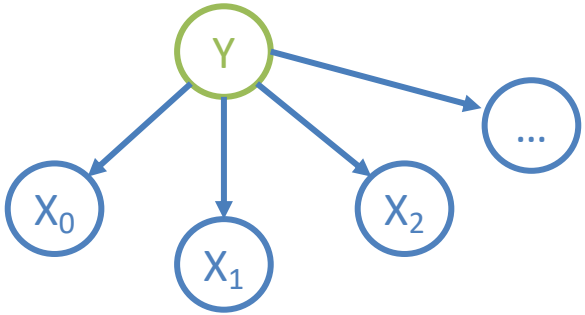
Announcements

- Classwork 5
  - Due **this Friday, 11/8 @5pm**
- Homework 3
  - Due next Friday, 11/15 @5pm

This Lecture

- ML Fundamentals
  - Overfitting
  - Smoothing
  - Tuning
  - Evaluation

Last Lecture



- Naïve Bayes Classifier (NBC)
  - Structure
    - Class Y is the parent of all feature nodes $X_i \in X$
  - Semantics
    - $P(X_i|Y) \forall X_i \in X$
  - Goal
    - Find the most likely y from X
    - $\hat{y} = argmax_{y \in Y} P(Y = y | X_0 = x_0, X_1 = x_1, \dots)$

Last Lecture

- Decision Rule

$$\hat{y} = argmax_{y \in Y} P(Y = y | X_0 = x_0, X_1 = x_1, \dots)$$
$$= argmax_{y \in Y} P(y) \prod_{x_i} P(x_i | y)$$
$$= argmax_{y \in Y} \log\big(P(y)\big) + \sum_{x_i} P(x_i | y)$$

- "Learning" from data
  - $P(x_i | y) = \frac{c(x_i)}{N} = \frac{\#\text{ occurance of } x_i}{\#\, samples}$

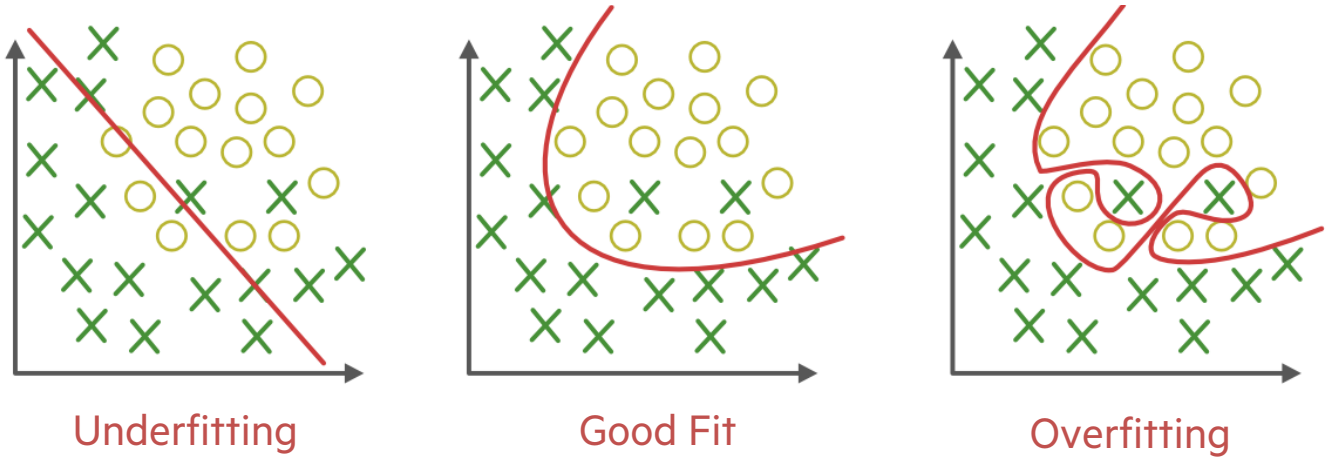
Aka Maximum Likelihood Estimation (MLE)

Overfitting

- Learning the training data **too perfectly**, including all its noise and random fluctuations, rather than learning the true underlying patterns.
  - Very high accuracy on training data
  - Poor performance on new, unseen data (validation/test sets)
- Examples
  - Memorizing exact answers to CW questions (training set) instead of understanding the concepts – failing on the exams (test set).

Overfitting vs. Underfitting

- Classification



Overfitting vs. Underfitting

- Regression



Overfitting in NBC

- NBC for spam detection
  - What may cause overfitting in NBC?
  - Limited Training Data
    - MLE estimation $P(x_i|y) = \frac{c(x_i)}{N}$ is less accurate
  - Zero Probability Problem: $P(x_i|y) = 0$ for word x_i in the test set that never appears in the training set
    - $\hat{y} = argmax_{y \in Y} P(y) \prod_{x_i} P(x_i|y) = 0 \; \forall y \in Y$

How to fix the Zero Probability Problem?

Smoothing

- Produce more **generalizable models** in machine learning by artificially inserting some information.
- Smoothing in NBC

$$P(x_i|Y = spam) = \frac{N_{x_i|spam} + \alpha}{N_{spam} + \alpha \cdot N_{vocab}}$$

$$P(x_i|Y = ham) = \frac{N_{x_i|ham} + \alpha}{N_{ham} + \alpha \cdot N_{vocab}}$$

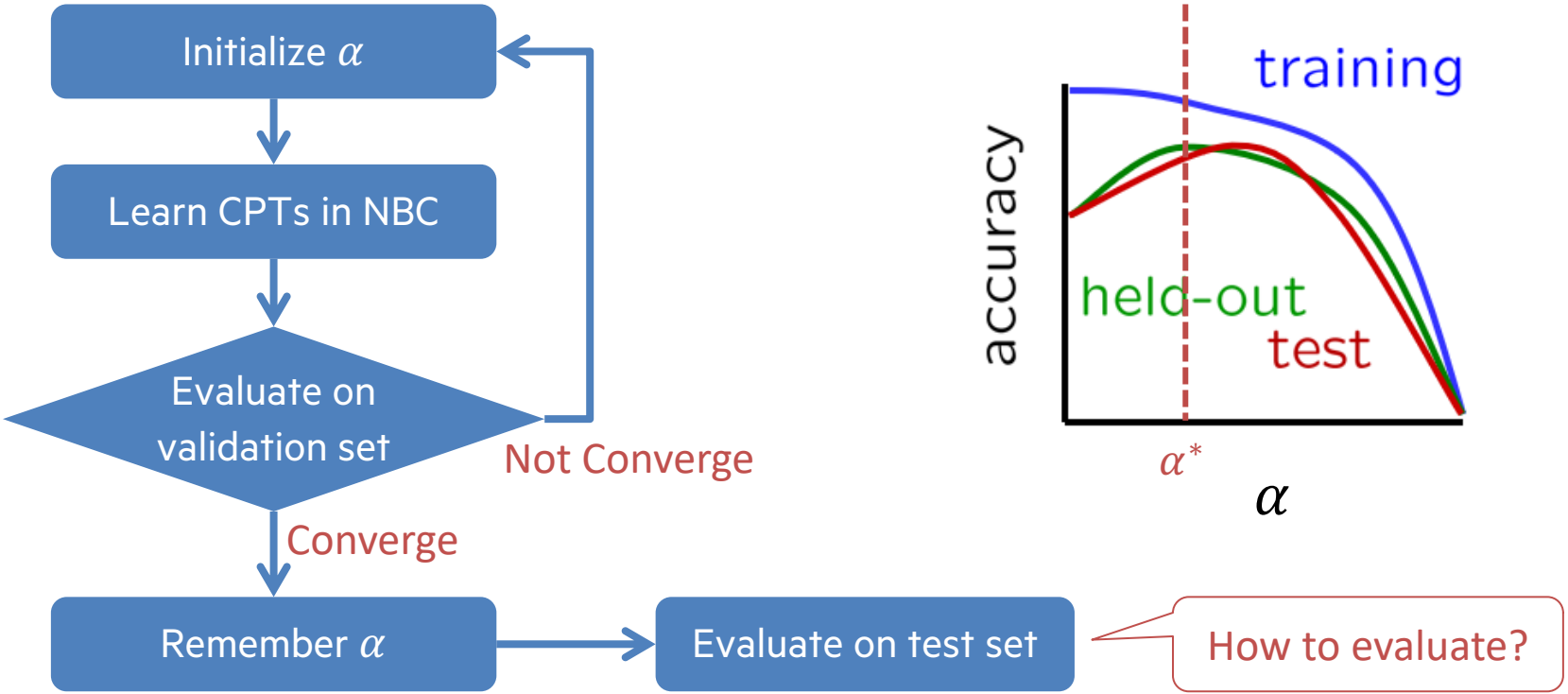
α is the **smoothing weight** to prevent zero probability problem/overfitting

How to determine the value of α?

Hyperparameter Tuning

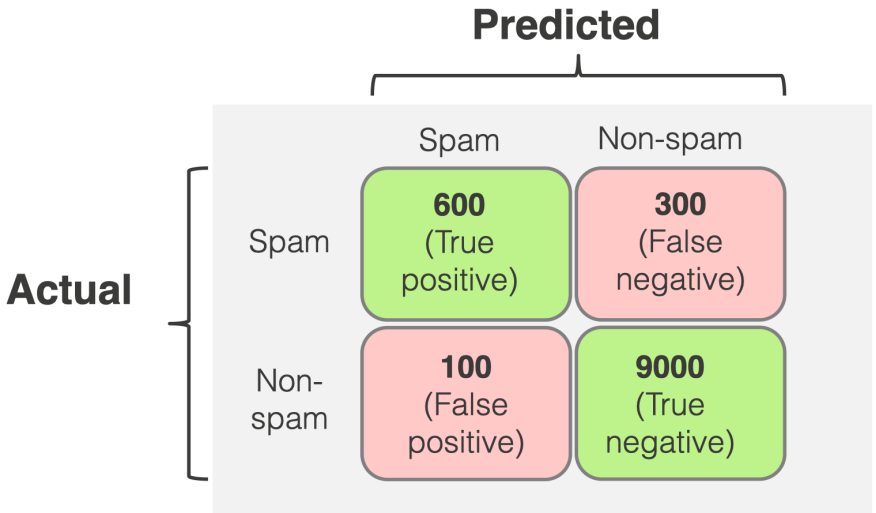
- Find the optimal values for a model's hyperparameters.
- **Parameters** and **Hyperparameters** in NBC
  - **Parameters**: automatically learned from the training data
    - CPTs: $P(y), P(x_i|y)$
  - **Hyperparameters** : manually tuned via searching/optimizing
    - Smoothing weight: α

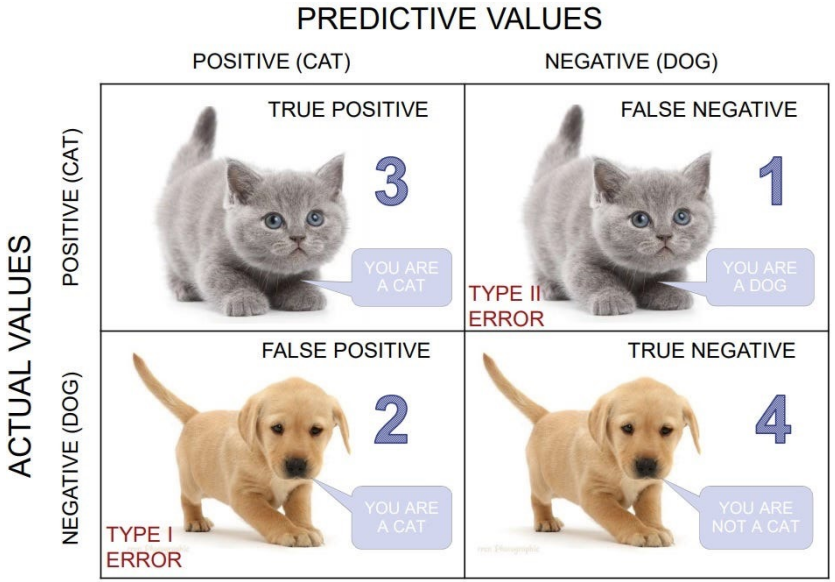
Hyperparameter Tuning



Evaluation – Classification

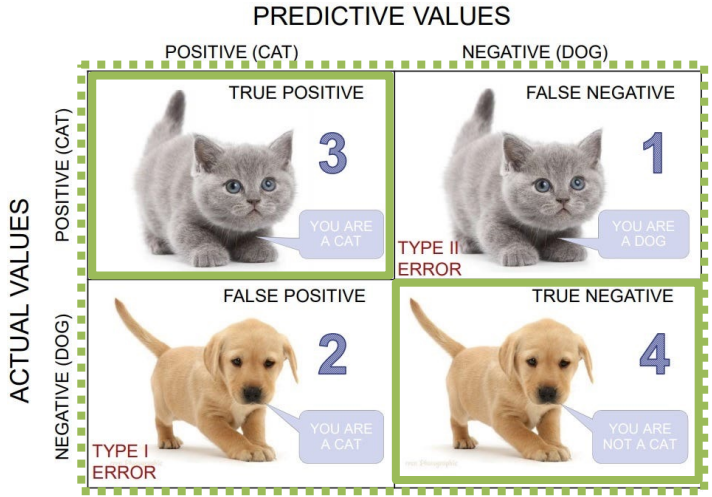
- Confusion Matrix





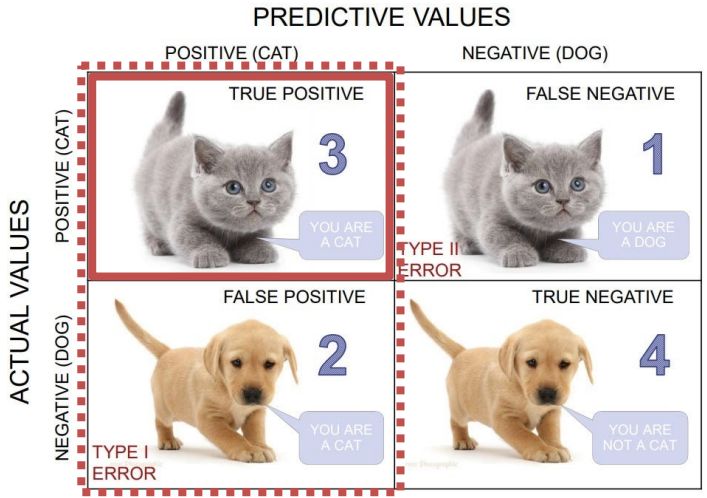
Evaluation – Classification

- Accuracy
  - % of correct predictions
  - $\frac{TP+TN}{TP+TN+FP+FN}$
  - Limitation
    - Treats all errors equally
      - Medical diagnosis: FN (missing a disease) could be life-threatening
    - Class imbalance problem
      - In a 99:1 dataset, 99% acc. may still miss all neg. samples



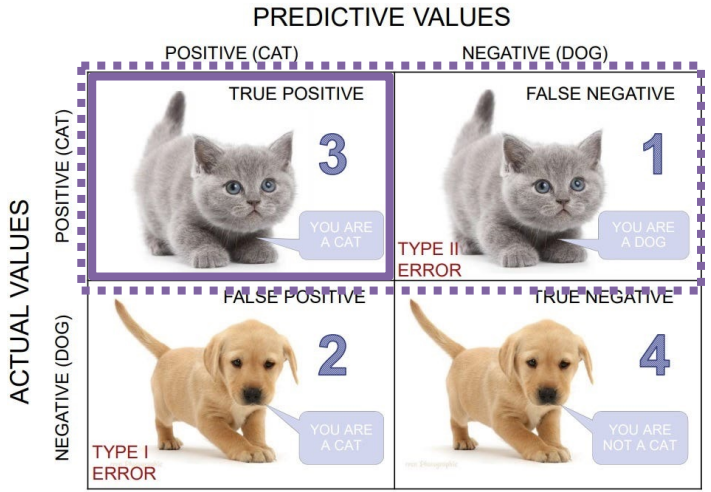
Evaluation – Classification

- Precision
  - % of true positives among positive predictions
  - $\frac{TP}{TP + FP}$
  - Informative when **false positives** are costly
  - E.g., important email wrongly marked as spam



Evaluation – Classification

- Recall
  - % of actual positives correctly identified
  - $\frac{TP}{TP + FN}$
  - Informative when **false negative** are costly
  - E.g., disease wrongly diagnosed as non-disease



Evaluation – Classification

- F1 -Score
  - Harmonic mean of precision and recall
  - $\frac{2*(Precision*Recall)}{Precision+Recall}$
  - Informative when the dataset is **imbalanced**
  - E.g., credit card fraud, rare disease

Evaluation – Regression

- Mean Squared Error (MSE)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

- Mean Absolute Error (MAE)

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

- R-squared (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

